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RADIOMETRIC AGES AND PALEOMAGNETISM OF THE SIGURA-GURA FORMATION, UPPER PART OF THE "TOBA TUFFS" IN SUMATRA, INDONESIA

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Abstract

Yokoyama, T., Dharma A. and Hehanussa, P., 1989. Radiometric ages and paleomagnetism of the Sigura-gura Formation, upper part of the "Toba Tuffs" in Sumatra, Indonesia. Palaeogeogr., Palaeoclimatol., Palaeoecol., 72: 161–175.

A radiometric age determination by means of K–Ar and fission-track method was carried out on the Sigura-gura Formation, upper part of the "Toba Tuffs" in Sumatra, Indonesia. The results are as follows: (1) K–Ar age: 0.96 ± 0.15 Ma. (2) Fission-track ages by re-etched method of external crystal surface are 1.11 ± 0.12 and 1.11 ± 0.14 Ma. (3) Fission-track ages by external detector method of crystal surface are 0.87 ± 0.09 and 0.86 ± 0.10 Ma. (4) The Paleomagnetic polarity of the Sigura-gura Formation is reversed. So this reversed polarity surely belongs to the Matuyama Reversed Epoch though it was previously correlated with the Blake Event in the Brunhes Normal Epoch. (5) The ash-flow deposit, which was the uppermost part of the Sigura-gura Formation, shows the youngest fission-track age, 0.03 ± 0.01 Ma in previous and this works. So, this must be a different unit, called the Porsea Formation, which overlies the Sigura-gura Formation. (6) The Porsea Formation may be a product of the same volcanic activity which erupted the Toba ash layer identified in deep sea sediments at about 0.075 Ma ago. (7) In addition, six fission-track age values were obtained from the Younger Toba Tuffs of "Toba Volcanic Complex". They are dated at 0.62–0.1 Ma.

Introduction

A large volume of acidic volcanic products, "The Toba Volcanic Complex" which was called "Toba Tuffs" previously, is distributed around Lake Toba, in northern Sumatra, Indonesia (Fig.1).

The "Toba Volcanic Complex" is composed of a thick sequence of pyroclastic flows, especially ash-flows, pumice-flows and welded tuff-deposits. According to the previous investigations (Yokoyama et al., 1980 and Yokoyama and Hehanussa, 1981), the so-called "Toba Tuffs", which we will call "Toba Volcanic Complex", do not comprise a single unit in the geologic sense, but judging from their composition and age, they can be divided into four different units as shown in Fig.2 and as follows:

(1) TukTuk Dacite: Mostly dense welded-tuff deposits with pumice flow deposits at the lowest and uppermost horizons. The age is about 1.9 Ma B.P. in the K–Ar age and the lower half of the Matuyama Reversed Chron in paleomagnetism.

(2) Older Toba Tuff: Mostly alternations of pyroclastic-flow and dense welded-tuff deposits. The age may be Late Matuyama Chron (1.2–1.3 Ma B.P.).

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(3) Younger Toba Tuffs: Mostly ash-flow deposits with dense welded parts at a few horizons. The age may be 0.1–0.7 Ma B.P. This unit can be divided into two units having regard to their origin from different eruption centers. One is the Sigura-gura Formation, and the other is the Haranggaol III and IV Formations.

(4) Samosir Formation: Lake sediments mostly exposed on Samosir Island.

In this stratigraphy, the reversed polarity of the Sigura-gura Formation in the "Younger Toba Tuffs" was correlated with the Blake Event in the Late Brunhes Normal Chron (Yokoyama and Dharma, 1984 and Nishimura et al., 1984), because of the fission-track age value, 0.1 ± 0.02 Ma, as reported by Nishimura (1980) and its reversed polarity.

But recent investigations by Yokoyama et al. (1985) reported on the age of the Sigura-gura Formation. In that report, Yokoyama et al. (1985) gave three fission-track age values for the uppermost and middle horizon of the Sigura-gura Formation and of the Porsea Formation taken as a part of the Sigura-gura Formation in previous reports. The two ages for the Sigura-gura Formation are 1.11 ± 0.12 (B Site; Loc.B.120) and 1.11 ± 0.14 (C Site; Loc.B.130) Ma, respectively. The two values given above were obtained by the re-etched and Grain-by-Grain Poisson-Test Calculation Method (GGPT Calculation Method: Yokoyama et al., 1984). In each case, these age data are discordant with the previous models.

In this report, we will present the new results of age determination and paleomag-

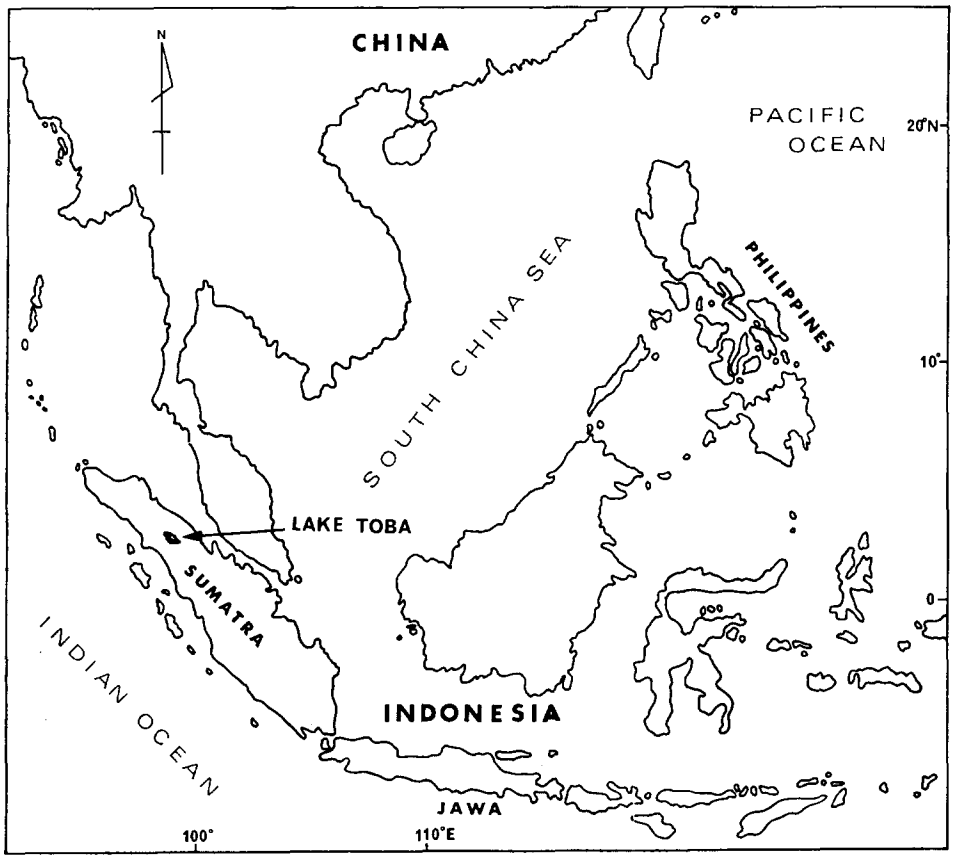


Fig.1. Location of Lake Toba, Sumatra.

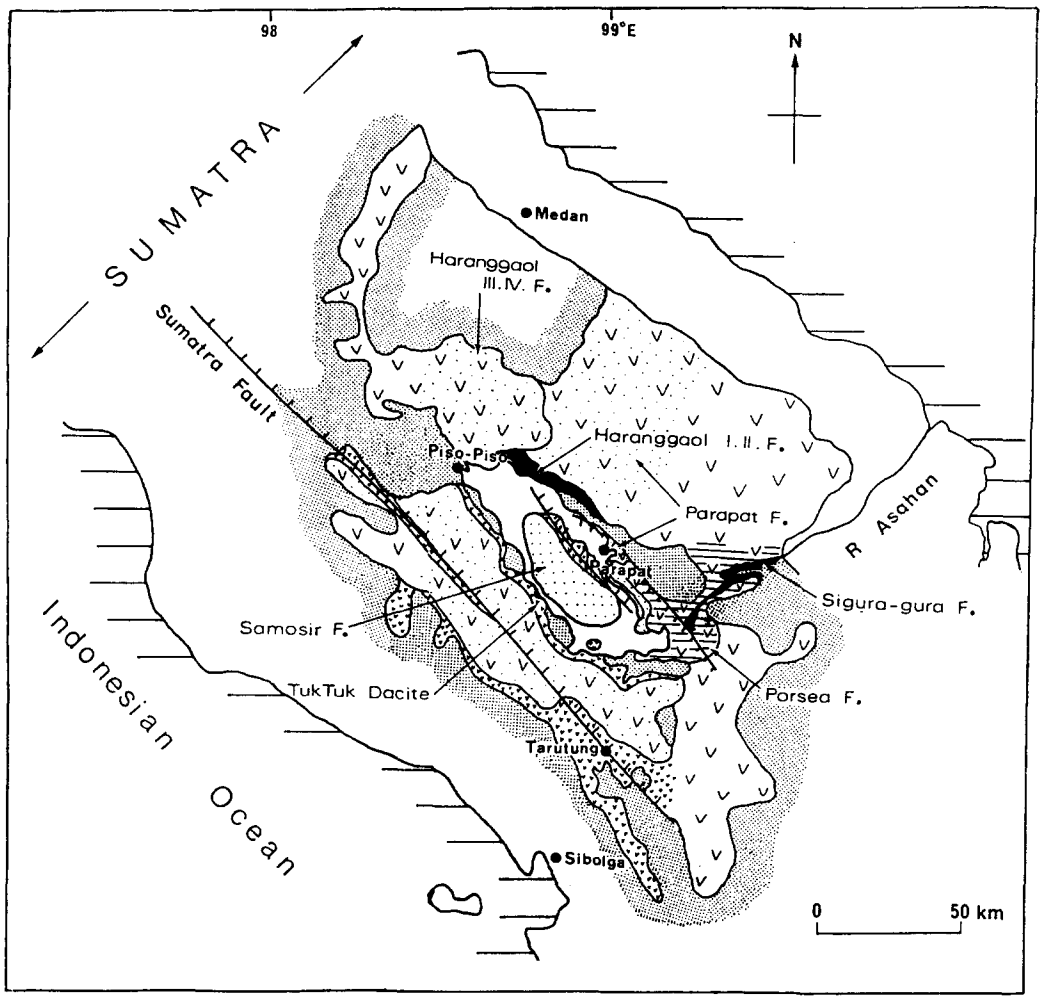


Fig.2. Geologic map of the "Toba Volcanic Complex".

netic measurements of the Sigura-gura Formation with some new age data on the "Toba Volcanic Complex", and will discuss the chronostratigraphic position of that formation within the "Toba Volcanic Complex".

Previous work on paleomagnetism of "Toba Tuffs"

(1) TukTuk Dacite: This body has both reversed and normal polarities. Situated on the west side of Lake Toba, the basal part shows flow structure and has a reversed polarity while the upper massive part has a normal polarity. The K–Ar age of 1.9 Ma B.P. was obtained by Tjia and Kusnaeny (1976) from samples near the basal part which coincides with the samples showing reversed polarity.

The paleomagnetic and age-dating data suggest that the normal polarity of this dacite body corresponds to the Olduvai Event in the Matuyama Reversed Epoch.

(2) Older Toba Tuff (Haranggaol I and II Formations): This is mainly found between Haranggaol and Parapat, along the northeast side of the lake. All parts of this formation showed a reversed polarity. Two age values were obtained from the basal part of this formation. One of them is the K–Ar age of 1.3 ± 0.4 Ma B.P. (Yokoyama and Hehanussa,

1981), and the other is a fission-track age of 1.2 ± 0.2 Ma B.P. (Nishimura et al., 1978a). The reversed polarity of the Old Toba Tuff certainly belongs to the Late Matuyama Reversed Chron.

(3) Younger Toba Tuff: Previously, this was divided into two units, i.e. the Haranggaol and the Sigura-gura Formations (Yokoyama and Hehanussa, 1981; Yokoyama and Dharma, 1984). The Haranggaol Formation has a normal polarity, which belongs to the Brunhes Normal Epoch.

Sampling and laboratory methods

Age determination at Porsea Area

From five sites around Porsea and Sigura-gura Fall, located as shown in Fig.3 (B.100–130 and D.140), six rock- and volcanic ash-samples were collected for age determination. In this paper, we will report the results from three sites and five specimens.

K–Ar age

Samples from site B.110 along the River Asahan were used for K–Ar age determination. This site (B.110) is in the wall of a newly excavated water tunnel of the Asahan Electric Power Plant, therefore the sample is very fresh. The measurements were carried out by Teledyne Isotope Co. Ltd. in U.S.A. (Table I).

Fission-track ages (Previous Data by the re-etched method, Yokoyama et al., 1985)

Samples from sites B.120, B.130 and D.140 were used for this method of dating. Site B.120 is in the river floor, about 300 m downstream from the Sigura-gura Fall, with locality number B. 120.

The dating method is outlined as follows: (a) zircon was separated from the specimens using the standard separating method with heavy solution and isodynamic separator. (b) After packing in an FEP sheet on a new-type heater, zircons were etched by 1:1, 48% HF and conc. H_2SO_4 in a stainless tank on the heater. After etching and washing the zircons were ready for counting. (c) Counting the number of spontaneous fission-track on the natural crystal surfaces under a microscope. (d) Re-etching under the same conditions as (b). (e) Identifying crystals in photographs and counting the induced fission-tracks on the same surfaces as the spontaneous fission-tracks. (f) Calculating the age of every zircon crystal and calculating the ages of the specimens by the GGPT Calculation Method (Yokoyama et al., 1984).

Fission-track ages (New data by external detector method: ED-2 Method)

The dating procedure is the same as the previous process up to step (c) as mentioned above. (d) Muscovite was used as a detector for irradiation in the reactor, using the external crystal surface.

(e) After identifying crystal position by photographs, the induced fission-tracks were counted on the same surfaces as the spontaneous fission-tracks. (f) Calculating the age of every zircon crystal and calculating the ages of the specimens by the GGPT Calculation Method.

Age determination of other Toba Volcanic Complex samples

The following 6 specimens were collected around Lake Toba from the Toba Volcanic Complex. The dating was carried out for these samples by the re-etched fission-track method.

Loc. 03; this specimen (PA-03) is from the outcrop in the Parapat area near Parapat Port facing Samosir Island.

Loc. 102: This sample (PH-102) is a pumice grain from the pumice flow deposit of the uppermost part of the Haranggaol IV Formation (see Yokoyama et al., 1980) at the northeastern side of Haranggaol.

Loc. 105: This is a specimen (PH-105) from a dense welded tuff bed of the Haranggaol III Formation (Yokoyama et al., 1980) at the northeastern side of Haranggaol. The stratigraphic position is just above the Matuyama/Brunhes Polarity boundary.

Loc. 115: This is a specimen (Sa-115) from

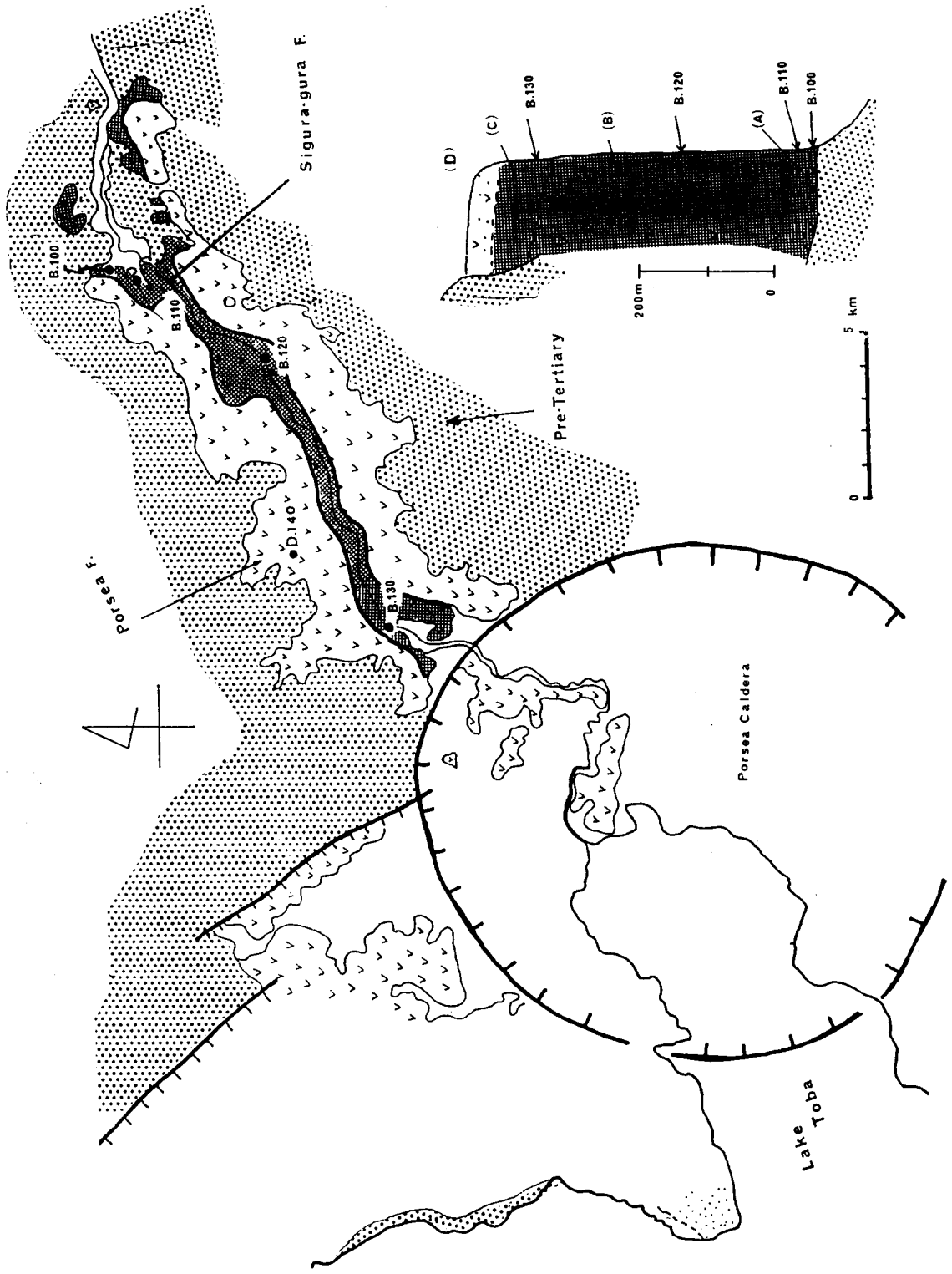


Fig.3. Geologic map of the Porsea-Asahan area around Sigura-gura Fall and sampling horizons and sites.

the thin tuff bed in the Samosir Formation at the western side of TukTuk Siadong on Samosir Island.

Loc. 1504: This is also from the Samosir Formation. The sample (PH-1504) consists of pumice grains contained in the sand layer of the Samosir Formation.

Loc. 1700: This sample (Tele-1700) is collected at Tele at the northern part of the lake.

Paleomagnetic measurements

From 4 sites (Loc. B. 100, 110, 120, 130), a total of 43 oriented samples were collected for paleomagnetic measurements. Three to ten samples were taken from each site, by drilling of several 1-inch cores. Remanent magnetization of the specimens was measured by a Digico spinner magnetometer.

Three specimens from each site were selected as pilot specimens for alternating field (AF) demagnetization tests. This was performed stepwise in peak fields up to 500 Oe. Most specimens did not show any significant change in paleomagnetic direction after AF demagnetization, with a few exceptions.

Using this procedure, the optimum demagnetizing field of the three pilot specimens was determined which minimizes the scatter of the magnetic directions and which shows the highest Fisher's precision parameter K. Then all other remaining specimens of the site were demagnetized in the selected optimum field.

Moreover thermal demagnetization experiment was carried out for these specimens from the Sigura-gura Formation (Torii et al., 1982). Three types of specimen were selected from the Sigura-gura Formation along Asahan River for this study. Type 1 is a densely welded, black, glassy tuff from the basal part of the formation (Site B. 110). Type 2 is a densely welded, white, devitrified tuff sampled at horizon of 50 m above the base (Site B. 120). Type 3 is a slightly weathered welded tuff collected from the uppermost part of the formation (Site B. 130). Though the vertical distance from type 1 to 3 is about 400 m, all specimens are presumed to belong to a single flow unit.

Results

The results of age determination are listed in Tables I–IV.

The results of paleomagnetic measurements are listed in Table V.

Discussion

Figure 4 represents a summary of the history of volcanic activity around Lake Toba. Using the results of the present work, the so-called "Toba Tuffs" can be divided into categories, as described below.

As was already stated in a previous report (Yokoyama and Hehanussa, 1981 and Yokoyama and Dharma, 1984), the TukTuk Dacite cannot be included in the category of the "Toba Tuffs", which represent a phase of volcanism which created Lake Toba. The TukTuk Dacite represents a phase of volcanism related to tectonic activities of the Sumatra Fault. Consequently, the term "Toba Tuffs" should be limited to the "Older and Younger Toba Tuffs".

Considering the results of the age determination, the Sigura-gura Formation must be a part

TABLE I

Results of K–Ar dating (Basal part of the Sigura-gura Formation)

Sample no.	Material analyzed	Isotonic age (Ma)	^{40}Ar (scc/gm 10^{-5})	$\%^{40}Ar$	%K
SIGA-1276	whole rock	0.96 ± 0.15	0.013	16.5	3.61
(Ka85-0625)			0.014	18.9	3.61

TABLE II

Results of fission-track dating (Re-etched method) of the specimens from B-D Beds of the Sigura-gura and Porsea Formation at Sites B.120, 130 and D.140

Grain number	Spontaneous number (N_s)	Fission-track density (ρs) ($\times 10^4/cm^2$)	Induced number (N_i)	Fission-track density (ρi) ($\times 10^5/cm^2$)	Measured square ($\times 10^{-5}$ cm^2)	Age (Ma)
Sigura-gura Formation (Lower part, (B)) [$\phi = 5.64 \times 10^{14}$ n/cm^2]						
1	3	2.24	136	10.14	13.41	0.74
2	2	4.93	19	4.68	4.06	3.54
3	3	4.27	45	6.40	7.03	2.24
4	1	2.80	31	8.68	3.57	1.08
5	4	4.64	78	9.05	8.62	1.72
6	0	0.00	29	5.91	4.91	0.00
7	3	6.16	55	11.29	4.87	1.83
8	1	2.29	95	21.74	4.37	0.35
9	2	6.02	46	13.86	3.32	1.46
10	5	5.27	121	12.76	9.48	1.39
11	4	12.86	41	13.18	3.11	3.28
12	1	2.94	26	7.65	3.40	1.29
13	2	4.42	32	7.06	4.53	2.10
14	2	5.62	43	12.08	3.56	1.56
15	1	3.75	42	15.73	2.67	0.80
16	2	1.77	151	13.33	11.33	0.45
17	1	2.98	28	8.33	3.36	1.20
18	3	3.09	125	12.89	9.70	0.81
19	3	4.69	59	9.22	6.40	1.71
20	2	6.04	50	15.11	3.31	1.34
21	5	12.53	164	41.10	3.99	1.02
22	4	11.46	56	16.05	3.49	2.40
23	3	11.03	41	15.07	2.72	2.46
24	9	12.86	188	26.86	7.00	1.61
25	2	5.21	62	16.15	3.84	1.08
26	8	17.06	164	34.97	4.69	1.64
27	4	3.46	370	32.03	11.55	0.36
28	6	8.93	105	15.63	6.72	1.92
29	1	1.32	96	12.72	7.55	0.35
30	7	7.43	131	13.91	9.42	1.80
31	4	3.75	48	4.50	10.66	2.80
32	1	0.99	75	7.45	10.07	0.45
33	4	6.09	92	14.00	6.57	1.46
34	5	7.85	104	16.33	6.37	1.62
35	6	5.87	308	30.14	10.22	0.65
36	4	7.21	72	12.97	5.55	1.87
Sigura-gura Formation (Uppermost Part, (C)) [$\phi = 5.65 \times 10^{14}$ n/cm^2]						
1	3	7.11	70	16.59	4.22	1.44
2	1	2.94	32	9.41	3.40	1.05
3	1	2.53	39	9.85	3.96	0.86
4	0	0.00	27	7.67	3.52	0.00
5	5	9.78	188	36.79	5.11	0.90
6	0	0.00	58	13.98	4.15	0.00
7	2	5.95	97	28.87	3.36	0.69
8	1	3.97	38	15.08	2.52	0.89
9	1	2.31	27	6.25	4.32	1.25
10	9	9.05	148	14.87	9.95	2.05
11	4	7.26	49	8.89	5.51	2.75
12	2	3.91	38	7.42	5.12	1.77
13	1	1.82	84	15.33	5.48	0.40

TABLE II *(continued)*

Grain number	Spontaneous number (N_s)	Fission-track density (ρs) ($\times 10^4/cm^2$)	Induced number (N_i)	Fission-track density (ρi) ($\times 10^5/cm^2$)	Measured square ($\times 10^{-5}$ cm^2)	Age (Ma)
14	2	3.75	40	7.50	5.33	1.68
15	1	3.61	13	4.69	2.77	2.59
16	4	8.18	58	11.86	4.89	2.32
17	4	10.00	40	10.00	4.00	3.37
18	8	10.97	103	14.13	7.29	2.62
19	4	7.72	36	6.95	5.18	3.74
20	1	1.44	93	13.36	6.96	0.36
21	2	1.05	59	3.11	19.00	1.14
22	0	0.00	35	4.70	7.44	0.00
23	2	5.36	41	10.99	3.73	1.64
24	1	1.82	59	10.77	5.48	0.57
25	4	6.94	170	29.51	5.76	0.79
26	3	4.97	54	8.94	6.04	1.87
27	5	5.57	168	18.71	8.98	1.00
28	3	9.06	31	9.37	3.31	3.26
29	2	1.18	120	7.06	17.00	0.56
30	0	0.00	14	3.98	3.52	0.00
31	4	9.43	51	12.03	4.24	2.64
32	2	4.81	70	16.83	4.16	0.96
33	3	4.16	113	15.65	7.22	0.89
34	2	4.85	47	11.41	4.12	1.43
35	8	22.54	181	50.99	3.55	1.49
36	1	1.73	87	15.08	5.77	0.39
Porsea Formation (D) [$\phi = 5.66 \times 10^{14}$ n/cm^2]						
1	0	0.00	121	24.15	5.01	0.00
2	0	0.00	143	35.84	3.99	0.00
3	0	0.00	49	11.26	4.35	0.00
4	0	0.00	111	21.02	5.28	0.00
5	0	0.00	150	15.08	9.95	0.00
6	0	0.00	61	15.06	4.05	0.00
7	0	0.00	817	35.51	23.01	0.00
8	0	0.00	339	17.84	19.00	0.00
9	0	0.00	216	15.06	14.34	0.00
10	0	0.00	71	7.02	10.11	0.00
11	1	1.77	61	10.78	5.66	0.55
12	0	0.00	173	15.20	11.38	0.00
13	0	0.00	78	10.76	7.25	0.00
14	0	0.00	38	8.54	4.45	0.00
15	0	0.00	57	13.35	4.27	0.00
16	0	0.00	39	9.38	4.16	0.00
17	0	0.00	85	5.67	15.00	0.00
18	0	0.00	69	8.28	8.33	0.00
19	0	0.00	106	8.83	12.01	0.00
20	0	0.00	142	15.32	9.22	0.00
21	0	0.00	88	8.92	9.86	0.00
22	0	0.00	111	20.00	5.55	0.00
23	0	0.00	126	55.02	2.29	0.00
24	2	0.71	545	19.46	28.01	0.12
25	0	0.00	170	23.91	7.11	0.00
26	0	0.00	64	11.68	5.48	0.00
27	0	0.00	38	13.38	2.84	0.00
28	0	0.00	39	7.69	5.07	0.00
29	0	0.00	116	10.55	11.00	0.00

TABLE II *(continued)*

Grain number	Spontaneous number (N_s)	Fission-track density (ρs) ($\times 10^4/cm^2$)	Induced number (N_i)	Fission-track density (ρi) ($\times 10^5/cm^2$)	Measured square ($\times 10^{-5}$ cm^2)	Age (Ma)
30	1	1.21	127	15.32	8.29	0.27
31	0	0.00	137	35.86	3.82	0.00
32	0	0.00	112	25.45	4.40	0.00
33	0	0.00	210	16.88	12.44	0.00
34	0	0.00	218	16.77	13.00	0.00
35	0	0.00	109	11.82	9.22	0.00
36	0	0.00	163	23.22	7.02	0.00

Specimen	Grain number (Σkn)	n	Theoretical		Obtained	kn	Others
			$fm(n)$	$fm(n) \times \Sigma kn$	$kn/\Sigma kn$		
Sigura-gura		0	0.069	2.5	0.111	4	
Formation (C)		1	0.185	6.7	0.250	9	
all grains		2	0.247	8.9	0.222	8	
		3	0.220	7.9	0.111	4	$m=2.667$
	36	4	0.146	5.3	0.167	6	$\Sigma N_s=96$
		5	0.078	2.8	0.056	2	$m/\sigma^2_{n-1}=0.530$
		6	0.035	1.2	0.000	0	
		7	0.013	0.5	0.000	0	
		8	0.004	0.2	0.056	2	
		9	0.002	0.1	0.028	1	
Sigura-gura		0	0.116	3.8	0.121	4	
Formation (C)		1	0.250	8.3	0.273	9	
without		2	0.269	8.9	0.242	8	$m=2.152$
no. 10, 18	33	3	0.193	6.4	0.121	4	$\Sigma N_s=71$
and 35		4	0.104	3.4	0.182	6	$m/\sigma^2_{n-1}=0.980$
		5	0.067	2.2	0.061	2	
Sigura-gura		0	0.038	1.4	0.028	1	
Formation (B)		1	0.124	4.5	0.194	7	
all grains		2	0.203	7.3	0.194	7	
		3	0.221	8.0	0.167	6	$m=3.278$
	36	4	0.181	6.5	0.194	7	$\Sigma N_s=118$
		5	0.119	4.3	0.083	3	$m/\sigma^2_{n-1}=0.739$
		6	0.065	2.3	0.056	2	
		7	0.030	1.1	0.028	1	
		8	0.012	0.4	0.028	1	
		9	0.007	0.2	0.028	1	
Sigura-gura		0	0.058	1.9	0.030	1	
Formation (B)		1	0.165	5.4	0.212	7	
without		2	0.235	7.8	0.212	7	$m=2.848$
no. 24, 26	33	3	0.223	7.4	0.182	6	$\Sigma N_s=94$
and 30		4	0.159	5.2	0.212	7	$m/\sigma^2_{n-1}=1.136$
		5	0.091	3.0	0.091	3	
		6	0.069	2.3	0.061	2	
Porsea		0	0.895	32.2	0.917	33	$m=0.111$
Formation	36	1	0.099	3.6	0.056	2	$\Sigma N_s=4$
		2	0.006	0.2	0.028	1	$m/\sigma^2_{n-1}=0.700$

$fm(n)$: Poisson Distribution $=(m^n/n!)\times e^{-m}$
k_n: Grain number having n spontaneous fission-tracks
m: Average of fission-track number
σ^2_{n-1}: Variance.

TABLE III

Results of fission-track dating (ED-2 Method: External detector method) of the specimens from B-D Beds of the Sigura-gura and Porsea Formations at Sites B.120, 130 and D.140 ($\lambda f = 7.03/10^{17}$y)

Grain number	Spontaneous number (N_s)	Fission-track density (ρs) ($\times 10^{-4}$/cm^2)	Induced number (N_i)	Fission-track density (ρi) ($\times 10^5$/cm^2)	Measured square ($\times 10^{-5}$ cm^2)	Age (Ma)
Sigura-gura B2 bed($\phi = 4.80 \times 10^{14}$/cm^2)						
1	2	7.22	77	2.77	27.80	0.74
2	4	14.44	83	2.77	29.96	1.38
3	4	14.44	209	2.77	75.45	0.55
4	5	21.65	89	2.31	38.53	1.61
5	4	27.03	138	1.48	93.24	0.83
6	2	9.05	40	2.21	18.10	1.43
7	2	7.22	71	2.77	25.63	0.81
8	3	20.27	54	1.48	36.49	1.59
9	5	18.05	69	2.77	24.91	2.07
10	12	43.32	308	2.77	111.19	1.11
11	1	6.76	33	1.48	22.30	0.87
12	1	6.76	84	1.48	56.74	0.34
13	1	7.75	30	1.29	23.26	0.95
14	0	0.00	48	1.38	34.78	0.00
15	0	0.00	bad condition	1.00		
16	3	16.30	49	1.84	26.63	1.75
17	1	7.75	43	1.29	33.33	0.67
18	4	14.44	125	2.77	45.13	0.92
19	1	3.39	93	2.95	31.53	0.31
20	10	38.76	146	2.58	56.59	1.96
21	1	3.61	47	2.77	16.97	0.61
22	3	11.63	70	2.58	27.13	1.23
23	5	19.38	153	2.58	59.30	0.93
24	0	0.00	24	1.11	21.62	0.00
25	3	11.63	133	2.58	51.55	0.65
26	2	7.22	108	2.77	38.99	0.53
27	0	0.00	72	1.38	52.17	0.00
28	1	13.51	25	0.74	33.78	1.14
29	1	4.33	83	2.31	35.93	0.34
30	0	0.00	130	2.77	46.93	0.00
31	2	9.05	141	2.21	63.80	0.41
32	3	21.74	70	1.38	50.72	1.23
33	1	10.87	36	0.92	39.13	0.79
34	3	21.74	137	1.38	99.28	0.63
35	4	14.44	108	2.77	38.99	1.06
36	2	21.74	43	0.92	46.74	1.33
Sigura-gura C bed ($\phi = 4.80 \times 10^{14}$/cm^2)						
1	2	10.87	101	1.84	5.49	0.57
2	3	16.30	84	1.84	4.57	1.03
3	1	5.43	46	1.84	2.50	0.62
4	1	5.43	62	1.84	3.37	0.46
5	9	32.49	290	2.77	10.47	0.89
6	4	14.44	93	2.77	3.36	1.24
7	0	0.00	43	1.66	2.59	0.00
8	2	10.87	65	1.84	3.53	0.88
9	0	0.00	40	1.29	3.10	0.00
10	2	7.22	91	2.77	3.29	0.63
11	1	5.43	96	1.84	5.22	0.30
12	2	14.49	34	1.38	2.46	1.69

TABLE III (*continued*)

Grain number	Spontaneous number (N_s)	Fission-track density (ρs) ($\times 10^{-4}$/cm^2)	Induced number (N_i)	Fission-track density (ρi) ($\times 10^5$/cm^2)	Measured square ($\times 10^{-5}$ cm^2)	Age (Ma)
13	0	0.00	67	1.94	3.45	0.00
14	0	0.00	bad condition	1.00		
15	3	16.30	77	1.84	5.22	0.30
16	4	20.62	163	1.94	8.40	0.70
17	6	21.66	124	2.77	4.48	1.39
18	5	18.05	131	2.77	4.73	1.10
19	0	0.00	56	1.84	3.04	0.00
20	5	25.77	118	1.94	6.08	1.22
21	2	7.22	51	2.77	1.84	1.13
22	2	7.22	124	2.77	4.48	0.46
23	3	15.46	85	1.94	4.38	1.01
24	1	3.61	77	2.77	2.78	0.37
25	1	3.61	41	2.77	1.48	0.70
26	3	20.27	64	1.48	4.32	1.35
27	1	3.61	62	2.77	2.24	0.46
28	2	10.87	56	1.84	3.04	1.03
29	1	3.61	51	2.77	1.84	0.56
30	2	7.22	67	2.77	2.42	0.86
31	6	32.61	102	1.84	5.54	1.69
32	4	21.74	70	1.84	3.80	1.64
33	0	0.00	60	1.84	3.26	0.00
34	2	10.87	67	1.84	3.64	0.86
35	1	9.01	29	1.11	2.61	0.99
36	5	18.05	98	2.77	3.54	1.47

of the Older Toba Tuffs. The fission-track age of the D Bed (D. 140) is very young, 0.03 Ma. This age is too young as obtained by the fission-track age dating method, therefore the age of 0.03 ± 0.01 Ma seems not so reliable. But it is certain that the D Bed is not a part of the Sigura-gura Formation. Therefore, it is named "Porsea Formation" in this paper.

The Sigura-gura Formation, which now consists of Beds A–C, is very thick. It reaches 600 m in thickness, but there is no indication of a stratigraphic boundary occurring in the body. It is considered that the Sigura-gura Formation is composed of one cooling unit and that it was erupted within a short time. As this formation filled-up a very narrow V-shaped valley, its volume is not so large. Yokoyama et al. (1980) supposed that the eruption center was certainly the Porsea Caldera, which was indicated by a gravity survey (Abe et al., 1980; Nishimura et al., 1984) and the distribution pattern of simple Bouguer anomalies. But this idea is not justified. Volcanic products originating from the Porsea Caldera belong to the D Bed, the Porsea Formation, which is the uppermost, loose part of the previous Sigura-gura Formation. Probably, this volcanic activity sent much volcanic ash into the Indian Ocean and on to the Malaya Peninsula, where material, produced by the same volcanic activity, was dated to be 75,000 yrs old by the K–Ar method by Ninkovich et al. (1978) and 30,000 yrs old by the fission-track method by Stauffer et al. (1980). However, these young age values obtained by fission track-dating usually have a big error range, as shown by only four spontaneous fission-tracks and the chemical similarity as reported by Stanffer et al. (1980).

TABLE IV

Results of fission-track dating from the samples of "Toba Volcanic Complex"

	Sample name										
	B2 (Re-etch)	B2 (ED-2)	C (Re-etch)	C (ED-2)	D (Re-etch)	PA-03 (Re-etch)	PH-102 (Re-etch)	PH-105 (Re-etch)	Sa-08 (Re-etch)	PH-1504 (Re-etch)	Tele-1700 (Re-etch)
Mineral	Zircon	Zircon	Zircon	Zircon	Zircon	Zircon	Zircon	Zircon	Zircon	Zircon	Zircon
Spontaneous track number (N_s)	94	96	71	86	4	12	84	35	12	8	12
Track density (ρs) (per cm^2)	4.60×10^4	1.33×10^5	3.83×10^4	1.15×10^5	0.13×10^4	4.02×10^3	3.26×10^3	1.80×10^4	1.29×10^4	3.86×10^3	7.06×10^3
Induced track number (N_i)	2845	3169	2146	2885	5299	3832	12929	1479	799	2069	1382
Track density (ρi) (per cm^2)	1.39×10^6	4.38×10^6	1.16×10^6	3.87×10^6	1.68×10^6	1.28×10^6	5.01×10^5	7.62×10^5	8.56×10^5	1.00×10^6	5.81×10^{14}
Flux value (ϕ) (per cm^2)	5.64×10^{14}	4.80×10^{14}	5.65×10^{14}	4.82×10^{14}	5.66×10^{14}	5.80×10^{14}	0.44×10^{15}	0.44×10^{15}	0.44×10^{15}	0.44×10^{15}	5.81×10^{14}
Age (Ma)	1.11 ± 0.12	0.87 ± 0.09	1.11 ± 0.14	0.86 ± 0.10	0.03 ± 0.01	0.11 ± 0.03	0.17 ± 0.02	0.62 ± 0.11	0.39 ± 0.11	0.10 ± 0.04	0.16 ± 0.05
Crystal no.	33	35	33	35	36	36	476	35	21	30	36
U-content (ppm)	123	456	102	401	148	111	57	87	97	140	132

TABLE V

Results of paleomagnetic measurements of the samples from the "Toba Volcanic Complex" containing the Sigura-gura Formation (Yokoyama et al., 1980 and Torii et al., 1982). 〈IS 105 and IS 102 are the same samples as PH-105 and PH-102 respectively in this report〉

Site	AF (Oe)	D (°)	I (°)	N	K	α95	VGP	
							Lat.	Long.
TukTuk Dacite								
IS. 604	100	191.3	−8.7	7	23.2	12.7	0.0	−78.5
IS. 607	100	17.8	−19.5	3	20.1	28.2	68.0	224.4
IS. 608	200	14.1	5.5	7	170.9	4.6	187.7	75.8
IS. 609	150	358.6	1.1	4	148.7	7.5	314.8	87.6
IS. 610	150	203.9	−3.7	14	83.0	14.5	9.7	−66.0
IS. 614	200	346.5	5.0	4	20.3	20.8	9.2	76.5
IS. 619	150	354.1	−18.3	5	14.1	21.0	304.4	76.5
IS. 501	300	205.5	−4.0	3	642.5	4.9	10.1	−64.5
IS. 502	50	193.6	−15.4	2	–	–	347.7	−75.8
Toba Tuff at Haranggaol								
IS. 101	100	0.6	0.1	12	43.2	6.7	87.1	266.3
IS. 103	50	355.8	4.9	11	173.9	3.5	85.8	3.1
IS. 104	50	0.7	4.2	13	85.2	4.5	88.9	234.5
IS. 105	300	359.4	6.2	6	499.3	3.0	89.4	30.0
IS. 106	150	185.1	−23.8	13	99.1	4.2	−79.2	306.2
IS. 107	200	178.3	−18.4	8	217.1	3.8	−83.2	264.6
IS. 108	300	183.5	−15.6	14	226.9	2.6	−83.9	312.9
IS. 109	300	153.0	9.5	6	5.8	30.4	−62.0	172.9
IS. 110	100	183.4	−13.0	8	91.8	5.8	−85.0	321.4
IS. 111	300	193.3	5.6	6	11.8	20.3	−75.6	31.7
Sigura-gura Formation								
A. 100	150	195.2	−13.0	10	61.2	6.2	25.9	−74.2
Bl. 110	150	189.5	−7.5	10	34.9	8.3	18.9	−80.4
Bu. 120	150	191.7	−4.3	12	43.0	6.7	17.6	−78.2
C. 130	150	187.9	−9.6	16	45.5	5.5	26.6	−81.8
B. 100	150	194.2	−10.8	5	63.3	9.6	357.3	−75.3
B. 110	250	191.3	−11.7	4	304.8	5.2	352.2	−78.1
B. 120	150	185.1	11.5	3	222.8	8.2	68.0	−80.2
B. 120	250	183.3	7.7	3	128.9	10.9	72.1	−80.8
B. 130	250	191.2	−1.2	10	310.6	2.7	18.9	−78.6
Samosir Formation								
IS. 612	100	11.5	20.1	3	46.4	18.3	76.1	154.0
IS. 613	150	0.4	−14.7	2	–	–	–	–
IS. 617	100	346.9	16.1	6	19.3	15.6	75.8	31.8
IS. 620	150	2.7	3.3	3	186.9	9.0	87.0	208.9
IS. 622	150	6.4	9.4	4	15.9	23.7	83.2	170.7

AF:Alternating field. D:Declination. I:Inclination.
N:Number of specimen. K:Fisher's precision parameter.

Summary

(1) New age data of the Sigura-gura Formation and other Toba Volcanic Complex members were obtained by K–Ar and fission-track dating as shown in Tables I and IV.

(2) Previously, the reversed paleomagnetic polarity of the Sigura-gura Formation was correlated with the Blake Event in the Brunhes Normal Chron. Based on new data, it must now be correlated to the Matuyama Reversed Chron.

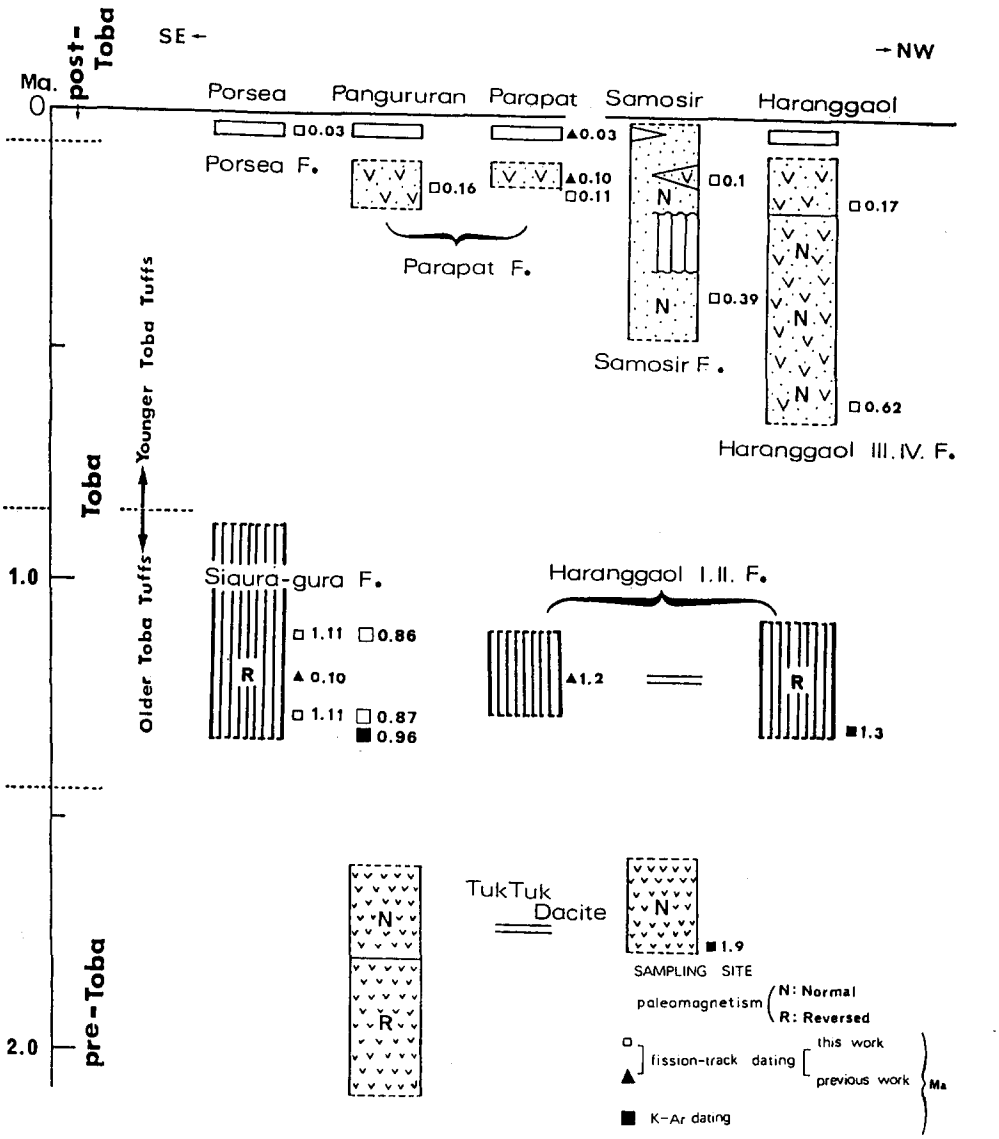


Fig.4. Volcano-stratigraphy of the "Toba Volcanic Complex" and the position of the Sigura-gura Formation.

(3) The Porsea Formation can be correlated with the ashes found in the Indian Ocean and on the Malaya Peninsula.

(4) A new model of the volcanic history of the Toba Volcanic Complex is constructed as shown in Fig.4.

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